

Integrated determinant witnesses and an explicit lower-end simplex stability power

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Outcome. For every fixed $d \geq 3$, every convex body $K \subset \mathbb{R}^d$, and every maximum-volume inscribed simplex S , we prove

$$E(K, S) \leq G_d^* \left(a(K) - \frac{1}{d+1} \right)^{1/d}$$

with entirely explicit constants. This removes the quadratic concentration loss of the released modulus; in dimension three the power is $1/3$ instead of $1/51$. The proof uses integrated determinant witnesses, a conditioned random anchor simplex, and the released elementary geometric conversion. It uses no inverse stability theorem for the Minkowski problem. No novelty, priority, best-known-rate, or optimal-exponent claim is made. An independent analytic model audit of this replacement passed. Finite exact checks support the audit; no human peer review or proof-assistant formalization is claimed.

1 Definitions and constants

For a full-dimensional compact convex body $Q \subset \mathbb{R}^n$, let ΠQ be its projection body, normalized by $h_{\Pi Q}(u) = |Q \mid u^\perp|_{n-1}$ for unit u , and let $R(Q) = |\Pi Q|/|Q|^{n-1}$. Entry005 defines the affine invariant

$$a(K) = \left(\frac{d}{d+1} \right)^d \frac{R(\mathcal{P}K)}{R(K)} - 1, \quad e = a(K) - \frac{1}{d+1} \geq 0,$$

where $\mathcal{P}K$ is a pyramid over K . For an inscribed simplex S with centroid z , the prescribed-center excess is

$$E(K, S) = \inf\{t \geq 0 : K \subset z + (1+t)(S-z)\}.$$

Fix an arbitrary maximum simplex S , and normalize it affinely to a regular simplex Δ of centroid zero and inradius one. Its vertex norms are d . The vertex-replacement determinant identity gives $|\alpha_i(x)| \leq 1$ for $x \in K$, and therefore

$$B_2^d \subset \Delta \subset K \subset R_0 B_2^d, \quad R_0 = d(d+1). \tag{1}$$

The resulting cone law ν is the pushforward of $h_K(u) dS_K(u)/(d|K|)$ by $u \mapsto u/h_K(u)$. It is centered, supported on $\partial K^\circ \subset B_2^d$, and satisfies

$$\begin{aligned} A &= \mathbb{E}|\det(X_1, \dots, X_d)| \leq 1, \\ B &= \mathbb{E}|\det((X_1, 1), \dots, (X_{d+1}, 1))|, \\ D &= B - A = (d+1)Ae \leq (d+1)e. \end{aligned}$$

These are the version-3 cone-law identities. With

$$b = \frac{1}{4(dR_0)^d}, \quad (2)$$

Cauchy's formula and the cube contained in the unit projected ball give

$$\mathbb{E}(-u \cdot X)_+ \geq 2b \quad (u \in S^{d-1}). \quad (3)$$

In detail the left side is $|K| |u^\perp| / (d|K|)$, which is at least $(2/d)^{d-1} / [d(2R_0)^d] = 2b$.

For this note define the explicit constants

$$Q = (d+1)(d+2)8^d b^{-4d}, \quad M = b^{-1}, \quad C = 4d^2 R_0(M+1), \quad (4)$$

$$e_* = \frac{1}{(d+1)MQ(8MdC)^d}, \quad A_d^* = 4MdC [MQ(d+1)]^{1/d}, \quad (5)$$

$$G_d^* = \max\{A_d^*, (R_0 - 1)e_*^{-1/d}\}. \quad (6)$$

All these constants are explicit. In particular b, Q, M, C, e_* are rational. They remain very poor numerically. No dimension-independent conclusion is claimed.

2 Integrated sign witnesses

We prove the new probabilistic step for any centered probability law on B_2^d satisfying (3); it need not be a cone law or have atoms. For an ordered base tuple $\mathbf{x} = (x_1, \dots, x_d)$ put

$$F_{\mathbf{x}}(y) = \det((x_1, 1), \dots, (x_d, 1), (y, 1)),$$

and let $P(\mathbf{x}) = \mathbb{E}(F_{\mathbf{x}}(X))_+$ and $N(\mathbf{x}) = \mathbb{E}(-F_{\mathbf{x}}(X))_+$. Centering gives the cancellation identity

$$D = 2\mathbb{E}_{\mathbf{X}} \min\{P(\mathbf{X}), N(\mathbf{X})\}. \quad (7)$$

Indeed $\mathbb{E}F_{\mathbf{x}}(X) = \det(x_1, \dots, x_d)$ up to an irrelevant orientation sign. Subtracting its absolute value from $\mathbb{E}|F_{\mathbf{x}}(X)|$ and then integrating proves (7), including singular base tuples.

Define a continuous, nonnegative two-sample witness

$$\psi_{\mathbf{x}}(y, z) = \min\{(F_{\mathbf{x}}(y))_+, (-F_{\mathbf{x}}(z))_+\} + \min\{(-F_{\mathbf{x}}(y))_+, (F_{\mathbf{x}}(z))_+\}. \quad (8)$$

For independent Y, Z of law ν , each minimum has expectation at most $\min(P, N)$, since it is bounded by each of its two arguments. Therefore

$$\mathbb{E}_{\mathbf{X}, Y, Z} \psi_{\mathbf{X}}(Y, Z) \leq D. \quad (9)$$

This integrates witnesses over their actual sampled points. It never assigns positive individual mass to a selected support point.

Lemma 1 (Linear assignment error). *There are $d+1$ affinely independent points $w_i \in \text{supp } \nu$ and a measurable assignment $X \mapsto w_{I(X)}$ such that*

$$h := \mathbb{E}\|X - w_{I(X)}\| \leq QD. \quad (10)$$

If $QD \leq b$, their simplex T satisfies $bB_2^d \subset T \subset B_2^d$.

Proof. Let $W = (W_0, \dots, W_d)$ be independent samples, and put

$$V(W) = |\det((W_0, 1), \dots, (W_d, 1))|.$$

For the covariance matrix $\Sigma = \mathbb{E}XX^\top$, (3) and centering give $\mathbb{E}|u \cdot X| \geq 4b$ and hence $u^\top \Sigma u \geq 16b^2$. Expanding the two determinants and using independence yields the exact moment identity

$$\mathbb{E}V^2 = (d+1)! \det \Sigma \geq (d+1)! 16^d b^{2d} \geq 2b^{2d}. \quad (11)$$

For clarity the lifted moment matrix is $\text{diag}(\Sigma, 1)$, and the usual determinant expansion gives $(d+1)!$ times its determinant. Also $V \leq 2^d$, by subtracting one lifted point from the others and applying Hadamard's inequality to the d difference columns. Thus the event

$$\mathcal{E} = \{V \geq v_0\}, \quad v_0 = b^d,$$

has probability at least

$$q_0 = \frac{b^{2d}}{4^d}, \quad (12)$$

because $\mathbb{E}V^2 \leq b^{2d} + 4^d \mathbb{P}(\mathcal{E})$.

For each i , use the d anchors except w_i as base, and test w_i and x in (8). For each $i < j$, use x and the $d-1$ anchors except w_i, w_j as base, and test w_i, w_j . Let $H(W)$ be the integral in x of the sum of all these $m = (d+1)(d+2)/2$ witnesses. Every summand has unconditional expectation at most D by (9): the base and the two tested samples are independent with law ν . Hence $\mathbb{E}H(W) \leq mD$.

Restricting to $\mathcal{E}$ and averaging, choose anchors in $\mathcal{E} \cap (\text{supp } \nu)^{d+1}$ with

$$H(W) \leq \frac{mD}{q_0}. \quad (13)$$

The inequality can be attained: H is continuous by dominated convergence, and the indicated event in the compact support product is compact and has positive probability. Its minimum does not exceed its conditional average. This selection argument is valid for nonatomic laws.

Let $\alpha_i(x)$ be the barycentric coordinates relative to these anchors. The first kind of witness is exactly $V \min\{1, (-\alpha_i(x))_+\}$. The second is at least $V \min\{(\alpha_i(x))_+, (\alpha_j(x))_+\}$: its two determinants have opposite signs whenever both coefficients are positive, and magnitudes $V\alpha_j(x)$ and $V\alpha_i(x)$. Contributions from two negative coefficients can be discarded. Consequently

$$\Phi := \mathbb{E} \left[\sum_i \min\{1, (-\alpha_i(X))_+\} + \sum_{i < j} \min\{(\alpha_i(X))_+, (\alpha_j(X))_+\} \right] \leq \frac{mD}{q_0 v_0}. \quad (14)$$

Every replacement determinant is at most 2^d , so

$$|\alpha_i(x)| \leq L_0 := 2^d/v_0 \geq 1 \quad (x \in B_2^d).$$

Assign x to the index r of its largest positive coefficient, resolving ties by the smallest index. Such a coefficient exists because their sum is one. Put $N_x = \sum_i (-\alpha_i(x))_+$ and $U_x = \sum_{i \neq r} (\alpha_i(x))_+$. Then

$$N_x \leq L_0 \sum_i \min\{1, (-\alpha_i(x))_+\}, \quad U_x \leq \sum_{i < j} \min\{(\alpha_i(x))_+, (\alpha_j(x))_+\}.$$

The second bound uses just the pairs containing r . Since the anchor diameter is at most two,

$$\|x - w_r\| \leq 2 \sum_{i \neq r} |\alpha_i(x)| = 2(N_x + U_x).$$

Integrating and using (14) gives

$$h \leq 2L_0\Phi \leq \frac{2L_0m}{q_0v_0}D = (d+1)(d+2)8^db^{-4d}D = QD.$$

Finally, for every unit u , the one-Lipschitz function $x \mapsto (u \cdot x)_+$ has expectation at least $2b$ by centering and (3). The assigned law has expectation at least $2b - h \geq b$. Some anchor therefore has $u \cdot w_i \geq b$. This holds in every direction, so $bB_2^d \subset T$. The other inclusion is automatic from the support condition. $\square$

3 Explicit geometric conversion

Now return to the cone law of the normalized body. If $e \leq e_*$, then $QD \leq Q(d+1)e \leq b$. Let $P = T^\circ$. Each selected w_i lies on ∂K° , so

$$K \subset P \subset MB_2^d. \quad (15)$$

The support function h_P is M -Lipschitz, $h_P(w_i) = 1$, and $h_K(x) = 1$ on $\text{supp } \nu$. Equation (10) gives

$$0 \leq \int (h_P(x) - 1) d\nu(x) \leq Mh \leq MQD =: z_0. \quad (16)$$

The left integral is $V(K[d-1], P)/|K| - 1$. Minkowski's first inequality therefore gives $(|P|/|K|)^{1/d} \leq 1 + z_0$. Here $z_0 \leq 1$, so

$$|P| - |K| \leq (2R_0)^d d 2^{d-1} z_0. \quad (17)$$

We recall the elementary cap argument to specify constants. Put $s = d_H(K, P)$. If $s > 0$, choose a farthest $q \in P$ and the unit direction u from its metric projection onto K , with $u \cdot q - h_K(u) = s$. We have $s \leq M$. With $t = s/[2(M+1)]$, the ball $(1-t)q + tB_2^d$ lies in P and its lowest u -coordinate exceeds $h_K(u)$ by at least $s/2$. It is disjoint from K . Its inscribed cube of side $2t/d$ gives

$$|P| - |K| \geq [s/(d(M+1))]^d.$$

Combining with (17), including $s = 0$, proves

$$s \leq Cz_0^{1/d} \leq C[MQ(d+1)e]^{1/d}. \quad (18)$$

Our gate e_* makes $s \leq 1/(8Md)$.

It remains to keep the initially chosen maximum simplex. Since $h_P \geq 1$ and $h_K \geq h_P - s$, we have $(1-s)P \subset K$. Maximality of Δ gives $|\Delta|/|P| \geq (1-s)^d \geq 1 - ds$. The nonnegative column-stochastic matrix of its vertex barycentric coordinates in P has determinant magnitude at least $1 - \delta$, where $\delta = ds \leq 1/8$. Hadamard's inequality forces every column to have an entry at least $1 - 2\delta$. The dominant rows are distinct: if two coincided, the permanent, interpreted as the probability of distinct independent row selections, would be at most $1 - (1 - 2\delta)^2 \leq 4\delta < 1 - \delta$, contradicting its domination of the determinant.

Match the vertices accordingly. Their distances are at most $2\delta \text{diam } P \leq 4Mds$. Taking convex combinations and using $B_2^d \subset \Delta$ gives

$$K \subset P \subset \Delta + 4MdsB_2^d \subset (1 + 4Mds)\Delta.$$

Together with (18) this proves $E(K, \Delta) \leq A_d^* e^{1/d}$ for $0 \leq e \leq e_*$. This includes $e = 0$: the assignment error and z_0 vanish, so $s = 0$ and the same prescribed simplex has zero excess. For $e > e_*$, (1) gives $E(K, \Delta) \leq R_0 - 1$, yielding (6). Undoing the affine map restores the centroid of the arbitrary initial S . We have proved:

Theorem 2 (Explicit unconditional replacement). *For every fixed $d \geq 3$, every convex body $K \subset \mathbb{R}^d$, and every maximum-volume inscribed simplex S , equations (2), (4)–(6) give*

$$E(K, S) \leq G_d^* \left(a(K) - \frac{1}{d+1} \right)^{1/d}.$$

At the local threshold, $A_d^* e_*^{1/d} = 1/2$.

4 Dependencies and the remaining geometric question

The new step is Lemma 1. It replaces shrinking-cell concentration by an integrated selection over conditioned random anchors. The cancellation identity, normalization, mixed-volume argument, cap estimate, and stochastic matrix conversion are standard or are already proved in the released supplement. No inverse-Minkowski result is used. The proof depends only on the direct arguments displayed here and the pinned cone-law definitions. No quantitative inverse-Minkowski theorem is imported.

The separate exact truncation calculation gives the ceiling $1/(d-1)$ for both the prescribed-maximum quantity and affine Banach–Mazur excess. It is not reproduced here. The remaining exponent loss is now solely the cap conversion from volume to distance: $1/d$ versus that ceiling. A valid comparison-simplex-specific inverse estimate might close it, but no such estimate is assumed or established by this note.

The exact checker supplements the proof by verifying affine determinant moments, cancellation and witness expectations, barycentric witness identities, and selection bounds for finite rational laws. Finite tests do not prove the general statement or cover nonatomic laws; those are handled analytically above. No novelty or best-known assertion is made.

References

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